

10 Challenge II

10.1 Work

Parameters: Below are the relevant parameters of the problem.

- **Site:** N is the number of production sites ($N = 30$). n is the index of production site, i.e. $n = 0$ refers to the first production site and $n = 29$ is the 30th.
- **Reseller:** M is the number of resellers, indexed by m from 0 to 3.
- **Time:** T is the time period, indexed by t , from 0 to 29.
- **Ingredient:** S is the number of ingredients, indexed by s , from 0 to 2, with 0 referring to sugar, 1 referring to powder milk, and 2 referring to flavour
- **Demand:** Each n production site has a daily demand for each product. This demand is $d_{n,t,s}$ referring to site, time, ingredient. Ex: $d_{20,3,2} = 20$ means the 21st site, in the fourth time period, has 20 units of demand for flavor.
- **Capacity:** Each reseller m has a capacity for each product s for each time period t $p_{m,s,t}$.
- **Inventory Cost:** Each n site has the ability to hold inventory at the end of a period t . We have an inventory cost $q_{n,s}$ as a parameter for the inventory holding cost associated with each site and ingredient.
- **Fixed Ordering Cost:** Each order has a fixed ordering cost parameter b_m that depends only on the reseller purchased from.
- **Variable Ordering Cost:** Each unit purchased as a cost c_s for each ingredient in the shipment, regardless of the seller.
- **Initial inventory:** We define an initial inventory for each site n and product s $v_{n,s}^0$

Variables: We can therefore create a set of variables to help us solve this problem.

- **Ordering Variable:** At each time step $t = 0, \dots, 29$, we must decide for each site $n = 0, \dots, 29$ how much of ingredient $s = 0, \dots, 2$ to order from each reseller $m = 0, \dots, 3$. We therefore create a variable $x_{n,m,t,s}$.
- **Inventory Stock:** We must also create a variable to denote, at the *beginning* of each time period t , how much inventory for each m ingredient to stock at each n site. This variable is $v_{n,t,s}$.
- **Fixed Order Binary:** $y_{n,m,t}$ to denote when we order from a manufacturer m to a site n in period t .

Objective: We want to minimize the total cost incurred by the company over the 30-day period while ensuring that each production site meets demand. This cost can be broken up into a few components.

$$\text{Inventory} + \text{Fixed Order Cost} + \text{Variable Cost} \quad (49)$$

Our variable cost will be constant, as we must meet demand over the course of each period, so we can exclude that from our problem. Inventory cost is the following equation:

$$\sum_N \sum_S q_{n,s} \sum_T v_{n,t,s} \quad (50)$$

In order to find the fixed order cost, we need to identify the number of orders we complete for each site n and seller m at each period t . We must create a binary variable $y_{n,m,t}$ to denote when we order from a manufacturer m to a site n in period t .

$$y_{n,m,t} = \begin{cases} 1 & \text{if } \exists s \in S \text{ such that } x_{n,m,t,s} > 0 \\ 0 & \text{otherwise} \end{cases} \quad (51)$$

The fixed order cost is then

$$\sum_M b_m \sum_N \sum_T y_{n,m,t} \quad (52)$$

Our final objective is therefore

$$\text{minimize } \sum_N \sum_S q_{n,s} \sum_T v_{n,t,s} + \sum_M b_m \sum_N \sum_T y_{n,m,t} \quad (53)$$

Constraints:

$$\text{(Inventory balance)} \quad (54)$$

$$v_{n,t+1,s} = v_{n,t,s} + \sum_{m \in M} x_{n,m,t,s} - d_{n,t,s} \quad \forall n, t, s \quad (55)$$

$$\text{(Reseller capacity)} \quad (56)$$

$$\sum_{n \in N} x_{n,m,t,s} \leq p_{m,s,t} \quad \forall m, t, s \quad (57)$$

$$\text{(Fixed cost linking)} \quad (58)$$

$$\sum_{s \in S} x_{n,m,t,s} \leq M_{n,m,t} y_{n,m,t} \quad \forall n, m, t \quad (59)$$

$$\text{(Binary variables)} \quad (60)$$

$$y_{n,m,t} \in \{0, 1\} \quad \forall n, m, t \quad (61)$$

$$\text{(Non-negativity)} \quad (62)$$

$$x_{n,m,t,s} \geq 0, \quad v_{n,t,s} \geq 0 \quad \forall n, m, t, s \quad (63)$$

10.2 Final Solution

$$\text{minimize} \quad \sum_{n \in N} \sum_{s \in S} \sum_{t \in T} q_{n,s} v_{n,t,s} + \sum_{m \in M} b_m \sum_{n \in N} \sum_{t \in T} y_{n,m,t} \quad (64)$$

$$\text{subject to} \quad v_{n,t+1,s} = v_{n,t,s} + \sum_{m \in M} x_{n,m,t,s} - d_{n,t,s} \quad \forall n \in N, t \in T, s \in S \quad (65)$$

$$v_{n,0,s} = v_{n,s}^0 \quad \forall n \in N, s \in S \quad (66)$$

$$\sum_{n \in N} x_{n,m,t,s} \leq p_{m,s,t} \quad \forall m \in M, t \in T, s \in S \quad (67)$$

$$\sum_{s \in S} x_{n,m,t,s} \leq M_{m,t} y_{n,m,t} \quad \forall n \in N, m \in M, t \in T \quad (68)$$

$$x_{n,m,t,s} \geq 0, v_{n,t,s} \geq 0 \quad \forall n, m, t, s \quad (69)$$

$$y_{n,m,t} \in \{0, 1\} \quad \forall n \in N, m \in M, t \in T \quad (70)$$